Hardwood Ballers Database

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INTRODUCTION

Introduction

- *What is the Hardwood Ballers Database?*
  - A collection of logically related data pertaining to the National Basketball Association (NBA)
  - Includes comprehensive information regarding player statistics as well as their associated positions, teams, personal attributes, schools and salaries

Introduction (continued)

- *What is the purpose of this database?*
  - To provide the ability to NBA fans as well as general individuals who are interested in assessing player performance with a multitude of different factors
  - Also delves deeper by allowing a user to gain insight on team performances relative to the players comprising them
  - Allows for the ability to adjust the provided attributes and update an individual player's information as well as creation of new ones

PRIMARY ENTITIES

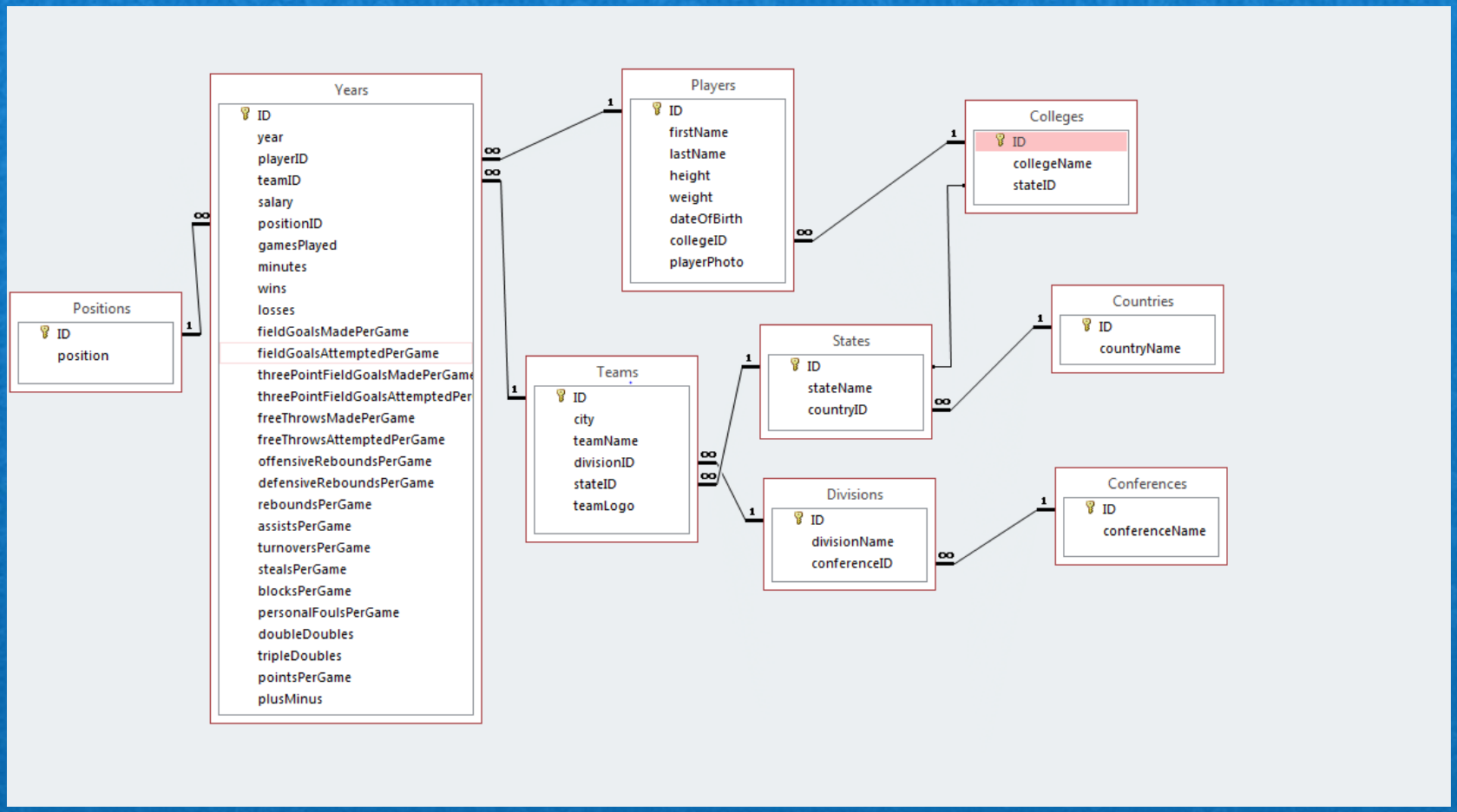
Primary Entity: Years

- Has 2281 rows of data with 27 attributes (excluding foreign keys)
- Provides detailed information of NBA players for the last 4 seasons (excluding the current season)
- Various statistics to measure player performance as well as salaries
- Data provided for statistical information (NBA.com) and salaries (basketball-reference.com and espn.com) was combined
- playerID and teamID were generated in Excel using VBA scripting

Primary Entity: Players

- Has 691 rows of data with 6 attributes (excluding foreign key)
- Provides personal information relating to height, weight, DOB of individual players
- For practical purposes in a report that we generated, we limited the "playerPhoto" attribute to the top 20 scorers of the 2012-2013 season
- collegeID was generated in Excel using VBA scripting

RDM


Years
ID
year
playerID
teamID
salary
positionID
gamesPlayed
minutes
wins
losses
fieldGoalsMadePerGame
fieldGoalsAttemptedPerGame
threePointFieldGoalsMadePerGame
threePointFieldGoalsAttemptedPer
freeThrowsMadePerGame
freeThrowsAttemptedPerGame
offensiveReboundsPerGame
defensiveReboundsPerGame
reboundsPerGame
assistsPerGame
turnoversPerGame
stealsPerGame
blocksPerGame
personalFoulsPerGame
doubleDoubles
tripleDoubles
pointsPerGame
plusMinus
Positions
ID
position
Players
ID
firstName
lastName
height
weight
dateOfBirth
collegeID
playerPhoto
Colleges
ID
collegeName
stateID
Teams
ID
city
teamName
divisionID
stateID
teamLogo
States
ID
stateName
countryID
Countries
ID
countryName
Divisions
ID
divisionName
conferenceID
Conferences
ID
conferenceName

Constraints

- Player age MUST be 19 years or older
- Salary has to greater than or equal to zero
- Games played has to be greater than or equal to zero
- Wins and losses have to be greater than or equal to zero
- All the statistic attributes have to be greater than or equal to zero (besides plus/minus)
- All the players who didn't go to college are international players or came straight from high school

QUERIES

John Hollinger's PER Query

A former ESPN NBA analyst and writer John Hollinger has developed a highly touted NBA statistic called the "Player Efficiency Rating" better known as PER. The PER provides a comprehensive statistic that encompasses all of a given player's contributions into one unique value.

```
SELECT Players.firstName, Players.lastName, Teams.city, Teams.teamName,
Years.year,
(((fieldGoalsMadePerGame*85.91)+(stealsPerGame*53.897)+(threePointFieldGoals
MadePerGame*51.757)+(freeThrowsMadePerGame*46.845)+(blocksPerGame*39.
19)+(offensiveReboundsPerGame*39.19)+(assistsPerGame*34.677)+(defensiveReb
oundsPerGame*14.707)-(personalFoulsPerGame*17.174)-
((freeThrowsAttemptedPerGame-freeThrowsMadePerGame)*20.091)-
((fieldGoalsAttemptedPerGame-fieldGoalsMadePerGame)*39.19)-
(turnoversPerGame*53.897))*(1/minutes)) AS PER
FROM Teams, Players, Years, Conferences, Divisions
WHERE Years.playerID = Players.ID AND Years.teamID = Teams.ID AND
Teams.divisionID = Divisions.ID AND Divisions.conferenceID = Conferences.ID
AND (minutes * gamesPlayed) >= 500 AND Years.year = EnterYear
ORDER BY Conferences.ID, Teams.city, Teams.teamName,
(((((fieldGoalsMadePerGame*85.91)+(stealsPerGame*53.897)+(threePointFieldGoa
lsMadePerGame*51.757)+(freeThrowsMadePerGame*46.845)+(blocksPerGame*39
.19)+(offensiveReboundsPerGame*39.19)+(assistsPerGame*34.677)+(defensiveRe
boundsPerGame*14.707)-(personalFoulsPerGame*17.174)-
((freeThrowsAttemptedPerGame-freeThrowsMadePerGame)*20.091)-
((fieldGoalsAttemptedPerGame-fieldGoalsMadePerGame)*39.19)-
(turnoversPerGame*53.897))*(1/minutes)))) DESC;
```

John Hollinger's PER Query

firstName	lastName	city	teamName	year	PER
Al	Horford	Atlanta	Hawks	2012	20.40
Josh	Smith	Atlanta	Hawks	2012	18.38
Jeff	Teague	Atlanta	Hawks	2012	17.26
Louis	Williams	Atlanta	Hawks	2012	16.42
Ivan	Johnson	Atlanta	Hawks	2012	16.32
Devin	Harris	Atlanta	Hawks	2012	15.08
Kyle	Korver	Atlanta	Hawks	2012	14.29
John	Jenkins	Atlanta	Hawks	2012	13.47
Zaza	Pachulia	Atlanta	Hawks	2012	13.46
Anthony	Tolliver	Atlanta	Hawks	2012	8.73
DeShawn	Stevenson	Atlanta	Hawks	2012	8.37
Dahntay	Jones	Atlanta	Hawks	2012	6.64
Kevin	Garnett	Boston	Celtics	2012	19.45
Paul	Pierce	Boston	Celtics	2012	19.30
Rajon	Rondo	Boston	Celtics	2012	18.23
Chris	Wilcox	Boston	Celtics	2012	15.45
Jeff	Green	Boston	Celtics	2012	15.15
Jordan	Crawford	Boston	Celtics	2012	13.99
Leandro	Barbosa	Boston	Celtics	2012	13.73
Jared	Sullinger	Boston	Celtics	2012	13.68
Jason	Terry	Boston	Celtics	2012	12.90
Brandon	Bass	Boston	Celtics	2012	12.50

Divisional Positions Salary Averages Query

An NBA team's General Manager is interested in seeing the salaries that are allocated to positional players in divisions in the past 4 years. Knowing so will enable the GM to know how much rival divisions are investing in specific positions.

Divisional Positions Salary Averages Query

```
SELECT Years.year, Divisions.divisionName, Positions.position, Avg(Years.salary) AS avgSalary
FROM Years, Teams, Divisions, Positions
WHERE (((Years.[teamID])=Teams.ID) AND ((Teams.divisionID)=Divisions.ID))
AND ((Years.positionID) =(Positions.ID)) AND Years.year = EnterYear
GROUP BY Years.year, Divisions.[divisionName], Positions.position;
```

Divisional Positions Salary Averages Query

year	divisionName	position	avgSalary
2010	Atlantic	Center	$3,153,849.10
2010	Atlantic	Point Guard	$3,536,053.56
2010	Atlantic	Power Forward	$5,029,809.90
2010	Atlantic	Shooting Guard	$3,553,931.71
2010	Atlantic	Small Forward	$4,563,837.76
2010	Central	Center	$3,669,441.38
2010	Central	Point Guard	$3,906,431.75
2010	Central	Power Forward	$3,558,294.86
2010	Central	Shooting Guard	$4,579,609.20
2010	Central	Small Forward	$5,187,701.50
2010	Northwest	Center	$5,183,022.67
2010	Northwest	Point Guard	$3,278,052.13
2010	Northwest	Power Forward	$4,588,307.75
2010	Northwest	Shooting Guard	$3,733,971.94
2010	Northwest	Small Forward	$4,334,617.92
2010	Pacific	Center	$6,727,168.33
2010	Pacific	Point Guard	$3,257,113.07
2010	Pacific	Power Forward	$3,036,067.19
2010	Pacific	Shooting Guard	$5,708,593.38
2010	Pacific	Small Forward	$3,337,211.00
2010	Southeast	Center	$4,184,302.16
2010	Southeast	Point Guard	$4,280,946.17
2010	Southeast	Power Forward	$5,372,579.44
2010	Southeast	Shooting Guard	$4,380,397.61
2010	Southeast	Small Forward	$4,102,683.32

True Shooting Percentage Query

This statistic provides a more in-depth perspective in a given player's overall shooting efficiency. It more accurately calculates a player's shooting than assessing FT%, FG%, 3-Point FG% individually. It calculates a value by combining FG% and 3-Point FG% into one unique FG% and then combining that with the FT%.

True Shooting Percentage Query

```
SELECT Players.firstName, Players.lastName, Teams.city, Teams.teamName,
Years.year, (pointsPerGame/(2*(fieldGoalsAttemptedPerGame+0.44*
freeThrowsAttemptedPerGame))) AS trueShootingPercentage
FROM Teams INNER JOIN (Players INNER JOIN Years ON Players.ID =
Years.playerID) ON Teams.ID = Years.teamID
WHERE ((([minutes]*[gamesPlayed])>=500) AND ((Years.year)= (Enter_Year)))
ORDER BY (pointsPerGame/(2*(fieldGoalsAttemptedPerGame+0.44*
freeThrowsAttemptedPerGame))) DESC;
```

True Shooting Percentage Query

firstName	lastName	city	teamName	year	trueShootingPercentage
Chris	Wilcox	Boston	Celtics	2012	0.725
Ryan	Hollins	Los Angeles	Clippers	2012	0.674
Tyson	Chandler	New York	Knicks	2012	0.671
Kevin	Durant	Oklahoma City	Thunder	2012	0.647
LeBron	James	Miami	Heat	2012	0.640
Kyle	Korver	Atlanta	Hawks	2012	0.637
Amar'e	Stoudemire	New York	Knicks	2012	0.637
Greg	Smith	Houston	Rockets	2012	0.636
Chris	Andersen	Miami	Heat	2012	0.631
Nick	Collison	Oklahoma City	Thunder	2012	0.624
Shane	Battier	Miami	Heat	2012	0.623
Jonas	Valanciunas	Toronto	Raptors	2012	0.619
Hasheem	Thabeet	Oklahoma City	Thunder	2012	0.619
Thabo	Sefolosha	Oklahoma City	Thunder	2012	0.617
Jose	Calderon	Detroit	Pistons	2012	0.616
Serge	Ibaka	Oklahoma City	Thunder	2012	0.612
Tiago	Splitter	San Antonio	Spurs	2012	0.609
Matt	Bonner	San Antonio	Spurs	2012	0.609
Kevin	Martin	Oklahoma City	Thunder	2012	0.608
Brandan	Wright	Dallas	Mavericks	2012	0.606
DeAndre	Jordan	Los Angeles	Clippers	2012	0.606
Carl	Landry	Golden State	Warriors	2012	0.605
Mickael	Gelabale	Minnesota	Timberwolves	2012	0.605
Steve	Nash	Los Angeles	Lakers	2012	0.605
Boris	Diaw	San Antonio	Spurs	2012	0.603
Steve	Novak	New York	Knicks	2012	0.602
Martell	Webster	Washington	Wizards	2012	0.601
Danny	Green	San Antonio	Spurs	2012	0.600
James	Harden	Houston	Rockets	2012	0.600
Ray	Allen	Miami	Heat	2012	0.599
Meyers	Leonard	Portland	Trail Blazers	2012	0.596
Mike	Miller	Miami	Heat	2012	0.596
Pablo	Prigioni	New York	Knicks	2012	0.595
Arnett	Moultrie	Philadelphia	Sixers	2012	0.595
Chris	Paul	Los Angeles	Clippers	2012	0.594
Kawhi	Leonard	San Antonio	Spurs	2012	0.592

“One-Two Punch” Query

An NBA analyst is intrigued to see the top scoring tandems on a given team are for each of the last four seasons. Finding this information will allow an analyst to use it for evaluation purposes such as the average points per game those two players account for of a team’s average points per game

"One-Two Punch" Query

```
SELECT year, firstName, lastName, teamName,
pointsPerGame
FROM Players, Years AS a, Teams AS t
WHERE a.teamID = t.ID AND a.playerID =
Players.ID AND a.ID IN (SELECT TOP 2 ID FROM
Years AS b WHERE b.year= a.year AND b.teamID =
t.ID ORDER BY pointsPerGame DESC)AND
Years.year = EnterYear
ORDER BY year, teamName, pointsPerGame DESC;
```

"One-Two Punch" Query

year	firstName	lastName	teamName	pointsPerGame
2010	Stephen	Jackson	Bobcats	18.50
2010	D.J.	Augustin	Bobcats	14.40
2010	Brandon	Jennings	Bucks	16.20
2010	John	Salmons	Bucks	14.00
2010	Derrick	Rose	Bulls	25.00
2010	Carlos	Boozer	Bulls	17.50
2010	Antawn	Jamison	Cavaliers	18.00
2010	JJ	Hickson	Cavaliers	13.80
2010	Paul	Pierce	Celtics	18.90
2010	Ray	Allen	Celtics	16.50
2010	Blake	Griffin	Clippers	22.50
2010	Eric	Gordon	Clippers	22.30
2010	Zach	Randolph	Grizzlies	20.10
2010	Rudy	Gay	Grizzlies	19.80
2010	Joe	Johnson	Hawks	18.20
2010	Josh	Smith	Hawks	16.50
2010	LeBron	James	Heat	26.70
2010	Dwyane	Wade	Heat	25.50
2010	David	West	Hornets	18.90
2010	Chris	Paul	Hornets	15.90
2010	Al	Jefferson	Jazz	18.60
2010	Paul	Millsap	Jazz	17.30
2010	Tyreke	Evans	Kings	17.80

College Average Salary Query

An NBA scout is interested in analyzing which colleges produce the highest valued players in terms of salary. This may provide insight for the scout on which colleges to prioritize his scouting assessments for.

College Average Salary Query

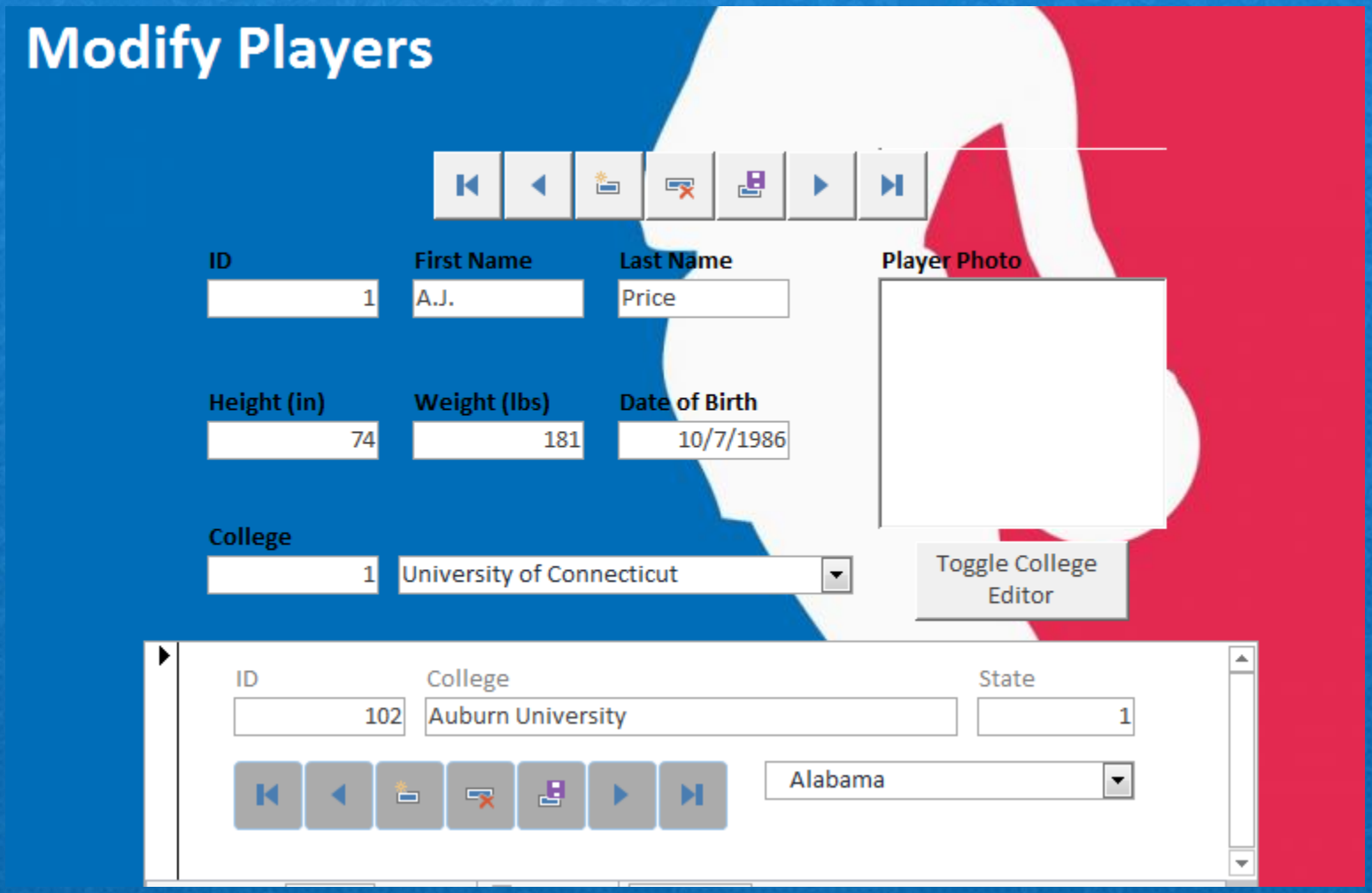
```
SELECT C1.collegeName, AVG (salary) AS [Average Salary], C2.N AS
[Number of Players]
FROM Colleges AS C1, Years, Players AS P,
     (SELECT collegeName, COUNT(T1.N) AS N
     FROM (SELECT DISTINCT Colleges.ID, Players.ID AS N
                   FROM Colleges, Players
                   WHERE Colleges.ID = Players.collegeID) AS T1,
                   Colleges AS C
     WHERE C.ID = T1.ID
     GROUP BY collegeName) AS C2
WHERE Years.playerID = P.ID AND P.collegeID = C1.ID AND
C1.collegeName = C2.collegeName AND Years.year BETWEEN StartYear
AND EndYear
GROUP BY C1.collegeName, C2.N;
```

College Average Salary Query

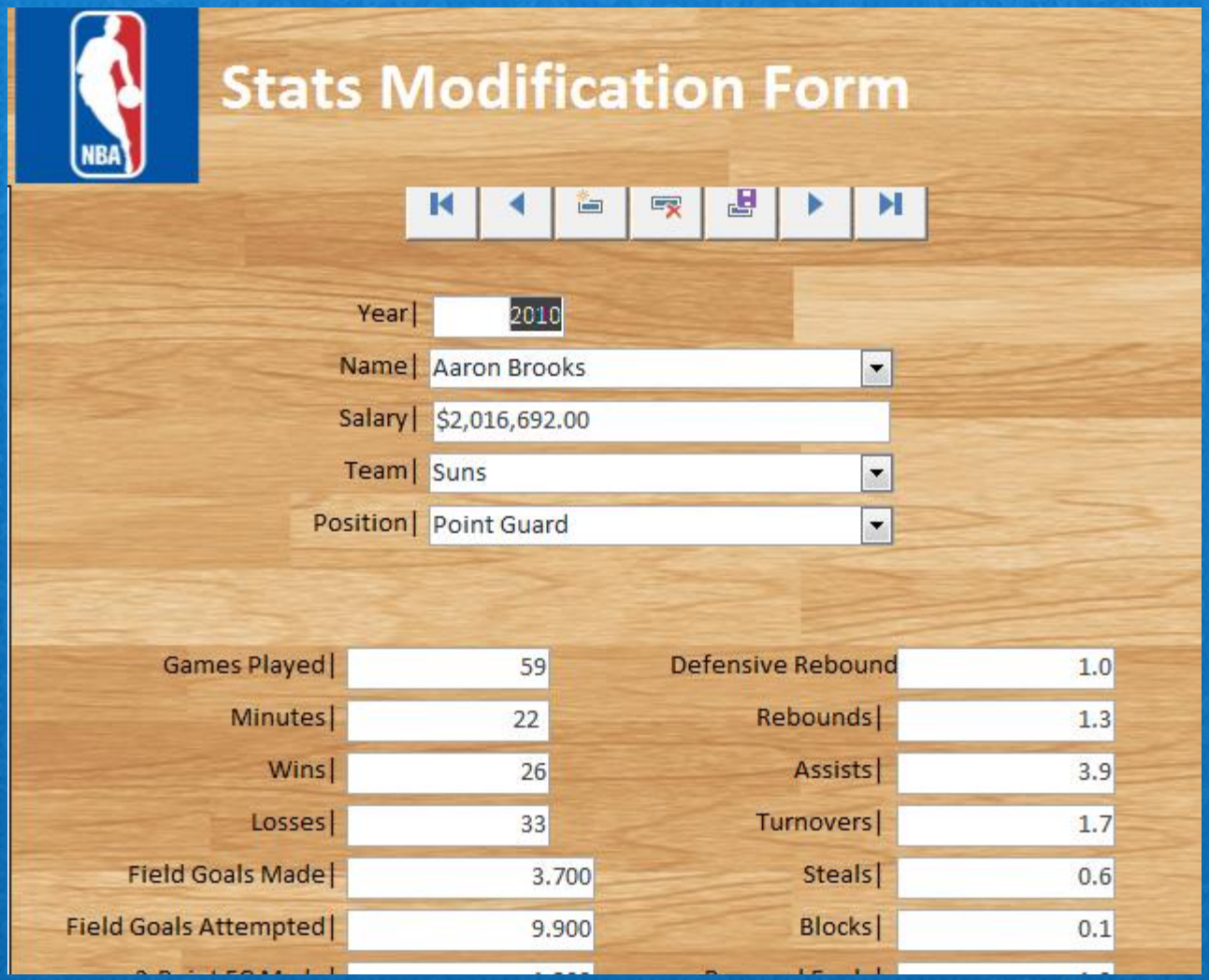
collegeName	Average Salary	Number of Players
Alabama A&M University	$137,498.00	1
Arizona State University	$2,555,271.50	4
Auburn University	$1,488,518.33	1
Augusta State University	$473,604.00	1
Baylor University	$1,680,359.00	5
Blinn College	$3,482,236.00	1
Boston College	$2,051,299.25	4
Bowling Green State University	$50,258.00	1
Bradley University	$2,552,884.50	1
Brigham Young University	$2,339,040.00	1
Butler County Community College	$6,524,250.00	1
Butler University	$1,670,497.00	2
California State University Fresno	$1,571,755.00	3
California State University Fullerton		1
Central Michigan University	$11,276,666.67	1
Clemson University	$1,616,766.67	1
Cleveland State University	$1,105,860.00	1
College of Charleston	$473,604.00	1
Colorado State University	$2,395,971.00	1
Cornell University	$38,194.00	1
Creighton University	$3,350,731.50	2

FORMS

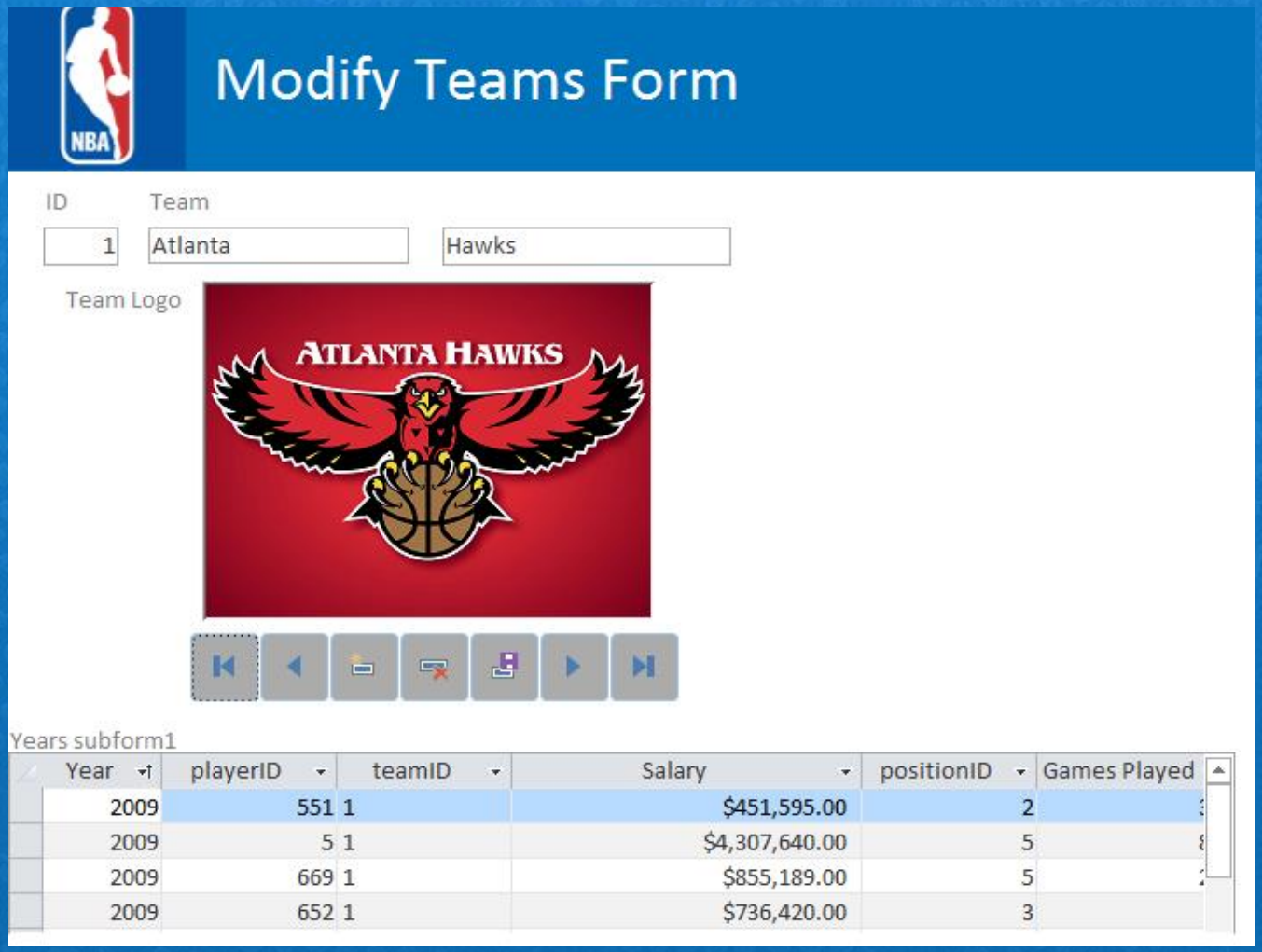
Modify Players Form



Stats Modification Form



Modify Teams Form



Year	playerID	teamID	Salary	positionID	Games Played
2009	551	1	$451,595.00	2	3
2009	5	1	$4,307,640.00	5	8
2009	669	1	$855,189.00	5	2
2009	652	1	$736,420.00	3	

REPORTS

Team Stats Report

- Our Team Stats report is for those who want to view all the players in each team of the years 2009-2012. The report lists the player's name, position, salary for specific years, and the six major statistics. It also displays the team's division and conference along with the total salaries paid and the number of players. This report is useful for NBA analysts or even basketball fans who want to analyze their team's player productivity.

Team Stats Report

Team Stats

2009-2012

Conference | Eastern

Division | Southeast

City | Atlanta

Team | Hawks

Total Salaries Paid | $274,481,947.00

Total Unique Players | 38

2009

First Name	Last Name	Position	Salary	Points	Rebounds	Assists	Steals	Blocks	Turnovers
Mike	Bibby	Point Guard	$6,217,617.00	9.10	2.3	3.9	0.8	0.0	1.1
Jason	Collins	Center	$825,497.00	0.70	0.6	0.2	0.1	0.1	0.2
Jamal	Crawford	Shooting Guard	$9,360,000.00	18.00	2.5	3.0	0.8	0.2	1.7
Maurice	Evans	Shooting Guard	$2,500,000.00	5.70	1.9	0.6	0.4	0.2	0.3
Al	Horford	Center	$4,307,640.00	14.20	9.9	2.3	0.7	1.1	1.5
Othello	Hunter	Small Forward	$736,420.00	1.60	1.7	0.0	0.0	0.1	0.4
Joe	Johnson	Shooting Guard	$14,976,754.00	21.30	4.6	4.9	1.1	0.1	1.9

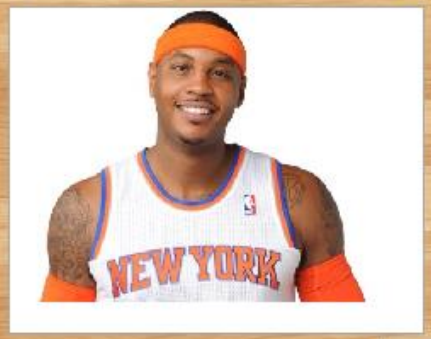
Top 20 Players Report (2012)

- This report is used to display the top 20 scorers in NBA from the 2012-2013 season. It displays the player's details and his major statistics such as points, rebounds, assists, steals, blocks, and the total games played for that season. This report can useful to the NBA analysts who want to compare the top 20 performers of the 2012-2013 season and evaluate their productivity to determine who should be considered the most valuable player.

Top 20 Players Report (2012)

Player ID
59

Player Name	Team Name
Carmelo Anthony	New York Knicks



Position
Small Forward

Points Per Game	Games Played	Rebounds/Game	Assists/Game	Steals/Game	Blocks/Game
28.66	67	6.9	2.6	0.8	0.5

Player ID
265

Player Name	Team Name
Kevin Durant	Oklahoma City Thunder